



1 million light-years away. This proved beyond doubt that the Universe was far larger than had previously been thought.

Investigating redshift

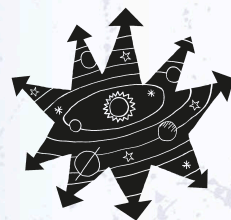
Redshift is a key concept in astronomy that was discovered by the American astronomer Vesto Slipher. He reported that spiral nebulae travel away from Earth after observing that the light they emit shifted toward the red end of the electromagnetic spectrum (rather than the blue end, which would indicate movement toward Earth). Hubble also examined the relationship between the distance of a galaxy from Earth and its redshift (when the light it emits increases in wavelength). He built on the research of other astronomers and verified Slipher's findings. He collected new data from other galaxies, which he plotted on a graph showing their distances relative to their redshifts.

Expanding Universe

Hubble's graph revealed a linear pattern, which indicated that the more distant the galaxy, the greater its redshift and therefore the faster it was receding from Earth. This phenomenon became known as Hubble's law. Hubble's surprising results, published in a landmark paper in 1929, made it clear that the existing notion of a stable and stationary Universe was incorrect and that we live in a large, dynamic Universe that is continually expanding in all directions. Since Hubble's discovery, scientists have drawn on his work to calculate the size of the observable Universe, to determine its age, and to estimate the moment of its origin.

**STUDIED THE
REDSHIFTS OF**

**46
GALAXIES**



**CALCULATED
ANDROMEDA
TO BE
900,000
LIGHT-
YEARS
AWAY**

**ESTIMATED
THE UNIVERSE
WAS AROUND
2 BILLION
YEARS OLD**